



Your environment, sensed. Its impact on you, explained.

An environmental health platform - a portable sensor, proprietary scores, and an AI companion for everything you can't see that affects how you feel and perform.

PRE-SEED · PERCH

2 · PROBLEM

Your environment is the largest untracked health metric - and the damage is documented.

We obsess over heart rate, sleep and steps. Meanwhile the environment we sit in impairs our driving, triggers our flares and migraines, and dulls our thinking - measurably, every day.

SAFETY

17.6%

of US fatal crashes involve a drowsy driver - an estimated 29,800 deaths over five years (AAA Foundation, 2017–2021).

- Cabin CO₂ passes 2,500 ppm within 15–20 minutes on recirculation, and exceeds 3,000 ppm in a full car with windows closed.
- In a 50-driver simulator study, levels above 1,400 ppm produced more traffic violations and measurably higher fatigue.
- Fatigue-related crashes cost the US roughly \$109B a year - and NHTSA considers drowsy driving heavily under-reported.

HEALTH

100M+

Americans live with conditions the environment can trigger - allergies, asthma, eczema, migraine, arthritis, MS (CDC, ACAAI, AMF, NEA).

- 39M Americans live with migraine; weather and air changes are among the most-cited triggers - reported by roughly half of patients (AMF, AAN).
- 60–80% of people with MS are heat-sensitive: small temperature rises temporarily worsen symptoms (Uhthoff's phenomenon).
- Sufferers keep symptom diaries - but the environmental variable is unmeasured, so triggers stay unidentified.

PERFORMANCE

90%

of our time is spent indoors, where pollutant concentrations run 2–5× higher than outdoor air (EPA).

- Decision-making and complex thinking measurably decline as CO₂ rises; effects reported below 2,000 ppm (Satish 2012; Allen 2015).
- Sealed bedrooms routinely exceed 2,000 ppm overnight - linked to lighter, more restless sleep.
- Offices reach 1,500+ ppm by mid-afternoon: the 3 p.m. brain fog is a measurable air problem, not a willpower problem.

All of it invisible. All of it measurable. None of it tracked.

Sources: AAA Foundation for Traffic Safety (2024); NHTSA; CDC; US EPA Indoor Air Quality & Asthma Triggers; WHO; Satish et al. (2012); Allen et al. (2015); peer-reviewed cabin-air studies (Hudda & Fruin; Jung et al.); ACAAI; American Migraine Foundation; National Eczema Association; NMSS/MSAA (Uhthoff's phenomenon); CDC NCHS Data Brief 460 (2021).

3 · SOLUTION

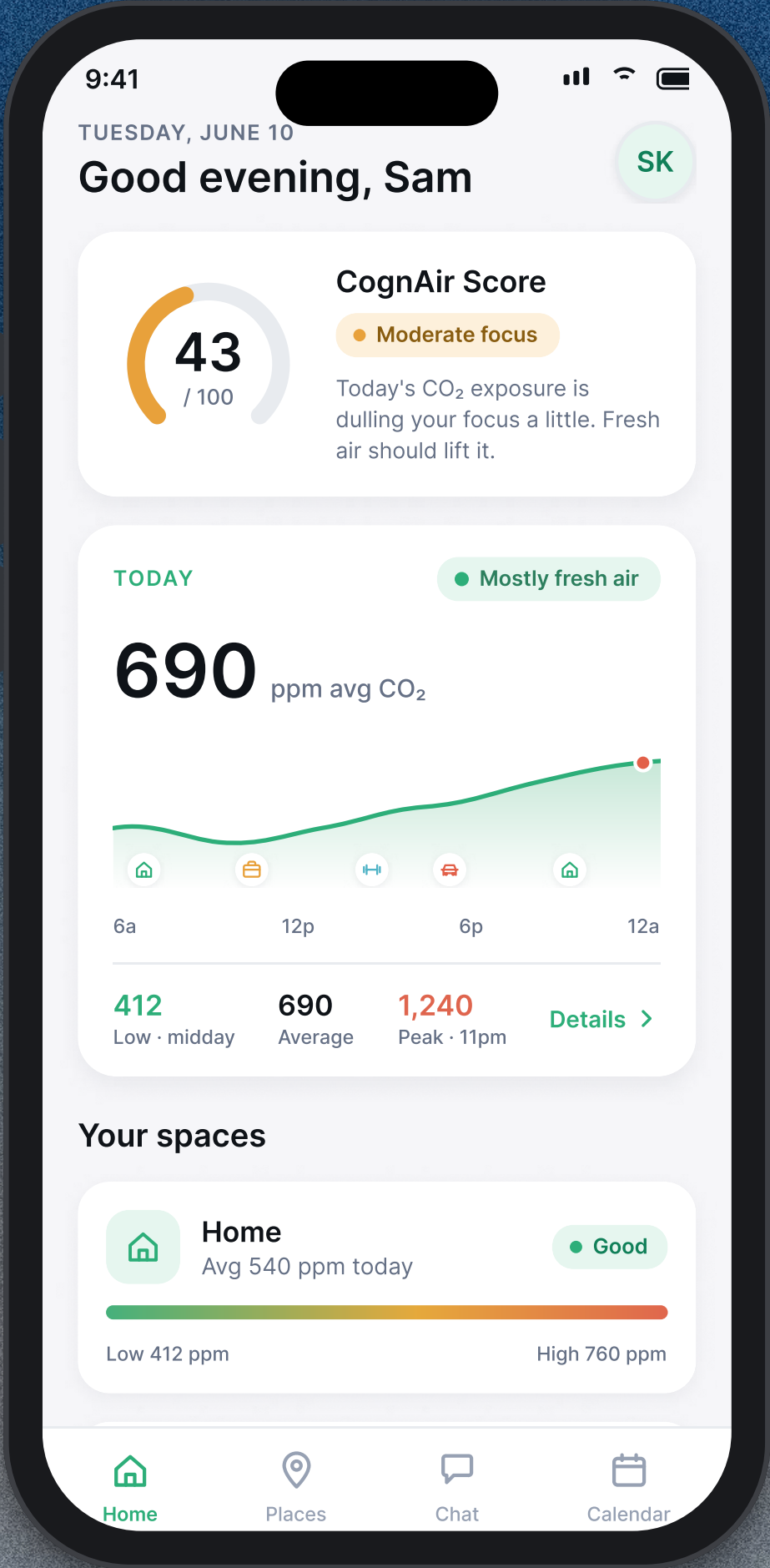
Meet Perch: track, score, and manage your environment.

- 1 Track**
Clip-on portable sensor follows you; GPS + BLE ties every reading to your spaces - bedroom, car, office, the morning run.
- 2 Score**
CognAir, SleepAir, FitAir and HealthAir turn raw air into research-based, patent-pending scores - proprietary to Perch Labs, and they personalize with use.
- 3 Manage**
Plain-language AI analyses and recommendations today; log symptoms and Perch cross-references your exposure to surface triggers; smart-home integrations tomorrow.

Oura and Whoop proved the playbook for physiological metrics - heart rate, HRV, sleep. Perch runs it for environmental metrics: CO₂, particulates, VOCs, NOx, temperature and humidity.



Perch Device



Perch App

4 · THE EXPOSURE MAP

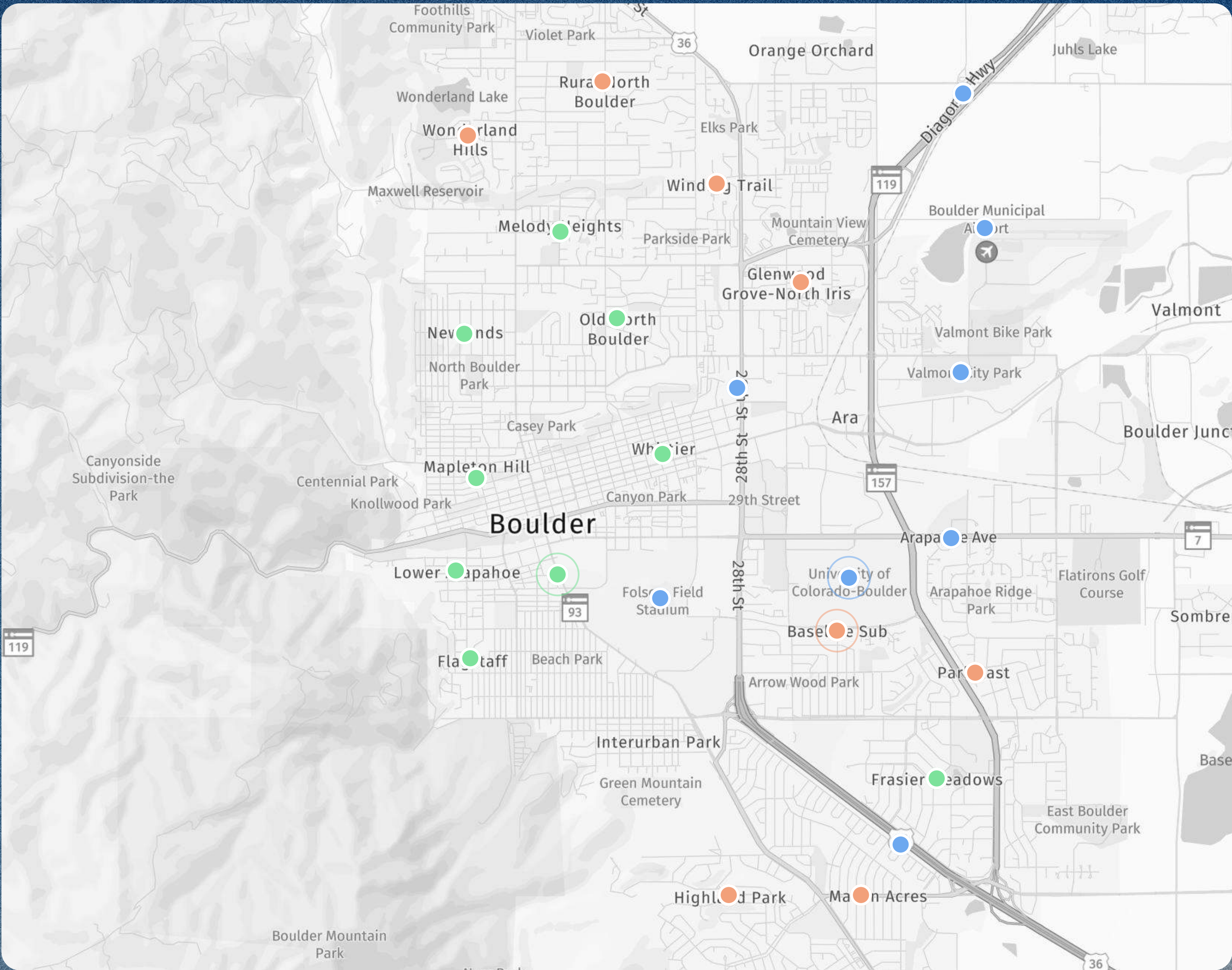
A few devices. A whole city, mapped.

Every clip-on records exposure wherever its member goes - bedrooms, commutes, offices, trails. The traces overlap, and a city's air comes into focus: block by block, hour by hour, indoors and out.

A few hundred members can blanket a city the size of Boulder - and every new device sharpens the map the data business is built on.

- Optimizers & sleep-hackers
- Trigger-condition members
- B2B fleets & workplaces

Each dot: one device, recording as its member moves.



Win one obsessive audience. Then widen.

NOW

Sleep & performance optimizers

Oura / Whoop communities, r/QuantifiedSelf, sleep-hacker creators. Channels: creator seeding, community launches, referral loops. They evangelize.

NEXT

Families & trigger-condition sufferers

Parents (nursery air), allergy, asthma, migraine, eczema and arthritis sufferers - symptom tracking + HealthAir are the key features. Channels: condition communities, paid social on proof content. Target HSA/FSA eligibility.

LATER

Fleets, offices & schools

Privacy-preserving alertness and IAQ compliance. Lighthouse pilots first; per-unit SaaS contracts follow.

Four curves just crossed.



Sensors got good and cheap

Sensirion's SEN66 packs PM, CO₂, VOC, NOx, temperature and humidity into one lab-grade module small enough to carry.



Wearables normalized daily scores

Oura and Whoop trained millions to check a daily readiness number every morning - the habit Perch's dataset is built on.



Air went mainstream

Post-pandemic ventilation awareness plus wildfire seasons made air quality a household concern.



AI closes the loop

AI models turn sensor streams into plain-language, personal recommendations - insight, not dashboards.

Working hardware. Working app. Patent-pending scores.

- Functioning portable prototype (Sensirion SCD-41) streams CO₂, temp and humidity over BLE - the production device moves to the Sensirion SEN66, adding PM, VOC, NOx and barometric pressure.
- GPS tags every reading to place; the platform builds your longitudinal exposure map.
- CognAir · SleepAir · FitAir · HealthAir distill it into scores - with one or two actions that matter today.

Alpha prototype pairing & syncing today



Perch app home screen



Perch alpha prototype

Hardware opens the door. The dataset is the business.

Device

\$179

One device - CO₂, particulates, VOC, NO_x, temperature, humidity and barometric pressure in every unit. Acquisition and daily presence - healthy margin, not the engine.

Perch Data

Licensing & API

Aggregated, privacy-first exposure data - indoors and out - licensed to HVAC & smart-home OEMs, insurers & real estate, and research institutions. Google paid \$200M+ for modeled outdoor data (BreezoMeter); Perch measures real exposure.

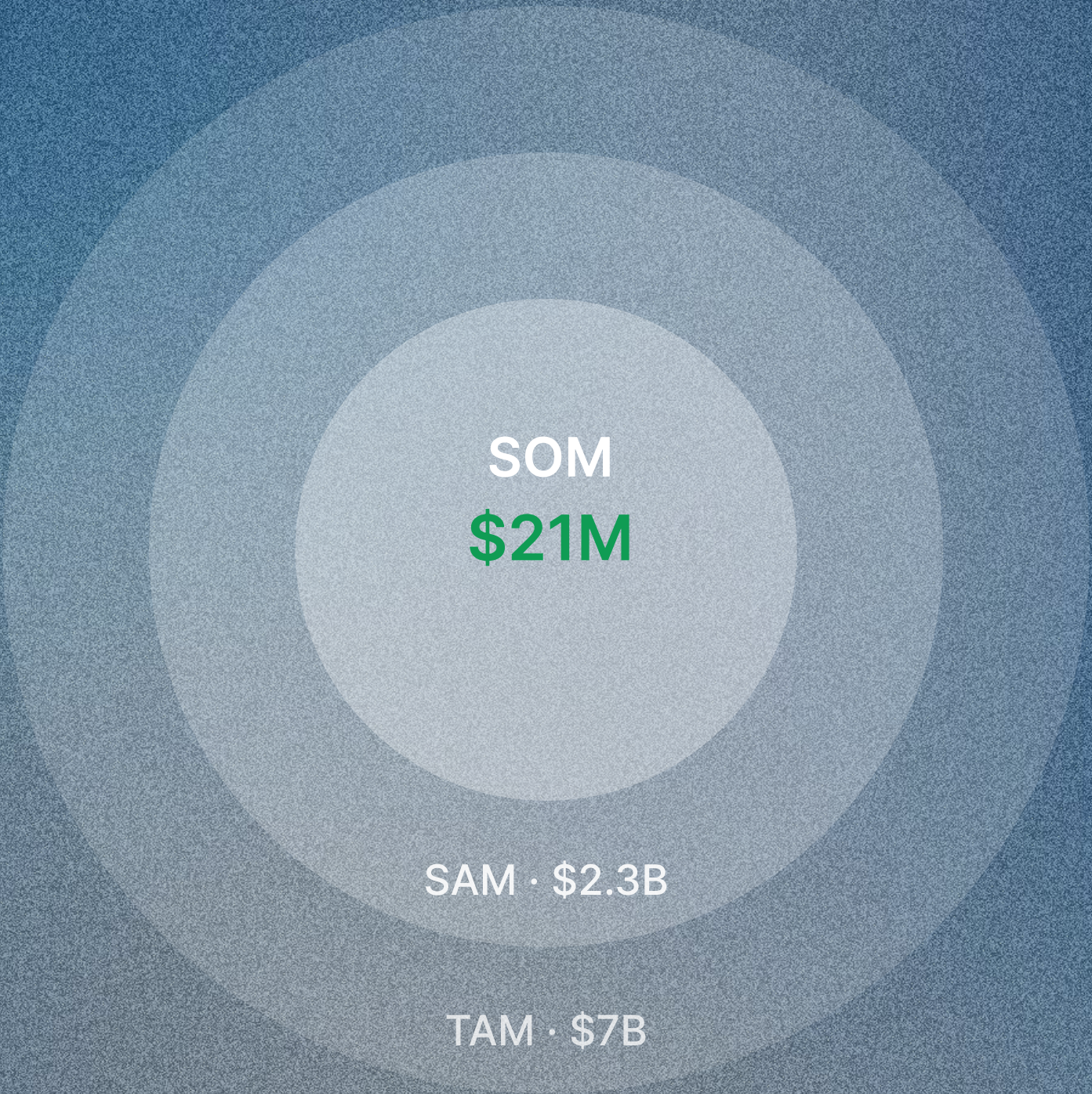
Fleet & Teams

\$15 / unit / mo

Per-vehicle and per-room licensing: dashboards, alerts, reporting. The enterprise expansion.

The app is free - scores, history and AI included - because every member compounds the dataset. Hardware carries margin today; data licensing is the recurring revenue engine tomorrow.

Everyone breathes. Start with the people who already track.



Beachhead

8M+ paying Oura & Whoop members already check a daily score; a \$179 device is a natural attach - and every member deepens the dataset.

Expansion

130M US households - bedrooms, nurseries and cars each justify a sensor; multi-device homes raise ARPU.

B2B

Fleets, offices and 130k US schools at \$15/vehicle-or-room/mo - the enterprise line.

Bottom-up SOM: 1.5% of the 8M-member Oura/Whoop base × \$179 device ≈ \$21M - before B2B and data licensing. TAM: global air-quality monitoring, \$6.8B (2025) → \$13B+ by 2035 (Emergen, Grand View). SAM: consumer indoor air monitoring, \$2.3B (2025, Fortune BI).

Two technical founders, a designer, a marketer, and a hardware engineer who's shipped.



Sam Kogan

CEO · CO-FOUNDER

Education: UCSD Math-CS
Built the Perch alpha end-to-end: sensor hardware, firmware, BLE sync and the app pipeline.



Amir Podoswa

COO · CO-FOUNDER

Education: CU Boulder
Neuroscience and Molecular Biology
Owns the science behind the scores, research partnerships and operations.



Noy Mizrachi

HEAD OF DESIGN

Education: Harvard GSD MDes.
Product, brand and the Perch app experience.



Devin Marcus

ELECTRICAL ENGINEER

Education: UCSD Electrical Engineering
Shipped consumer hardware. Guiding the alpha today; leads the SEN66 production device and DFM post-raise.



Aaron Goldenberg

COMMUNICATIONS & MARKETING

Education: UF Communication
Brand voice, community and launch - tells the Perch story to members, partners and press.

Environmental-sensing evangelists by necessity: trigger conditions run in our families - and in some of us. We're building the tool we wished existed. The raise brings the whole team on full-time.

Working alpha. \$1.5M makes us seed-ready.

PRODUCT PROGRESS

- ✓ Working alpha: SCD-41 device pairing & syncing with the app
- ✓ Android dev build running the core experience
- ✓ Production device (SEN66) spec'd on paper - sensor selected

MILESTONES · 18 MONTHS

M1

SEN66 prototype

first build, post-raise

M2

Campus betas

100+ users · CU Boulder, UCSD

M3

Data deals

campuses pay first

THE ASK

\$1.5M pre-seed

18 months of runway to:

The whole team full-time + a founding engineer	60%
SEN66 device prototype + hardware iteration	25%
Campus betas (100+ users), ops, patents & legal	15%

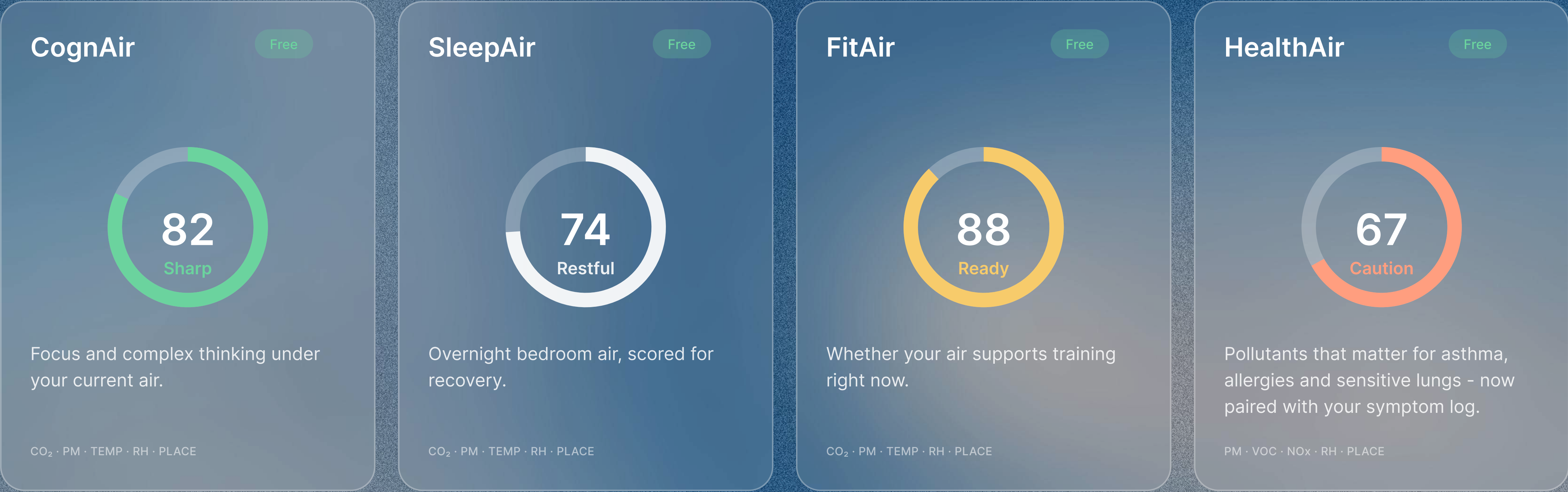
APPENDIX

The deep dives.

Scores, science, hardware, competition and roadmap - the detail behind the ten slides.



Research-based, patent-pending scores for every way air touches you.



Every Perch senses PM, VOC and NO_x - they sharpen CognAir, SleepAir and FitAir, and power HealthAir.

One answer instead of four readings.



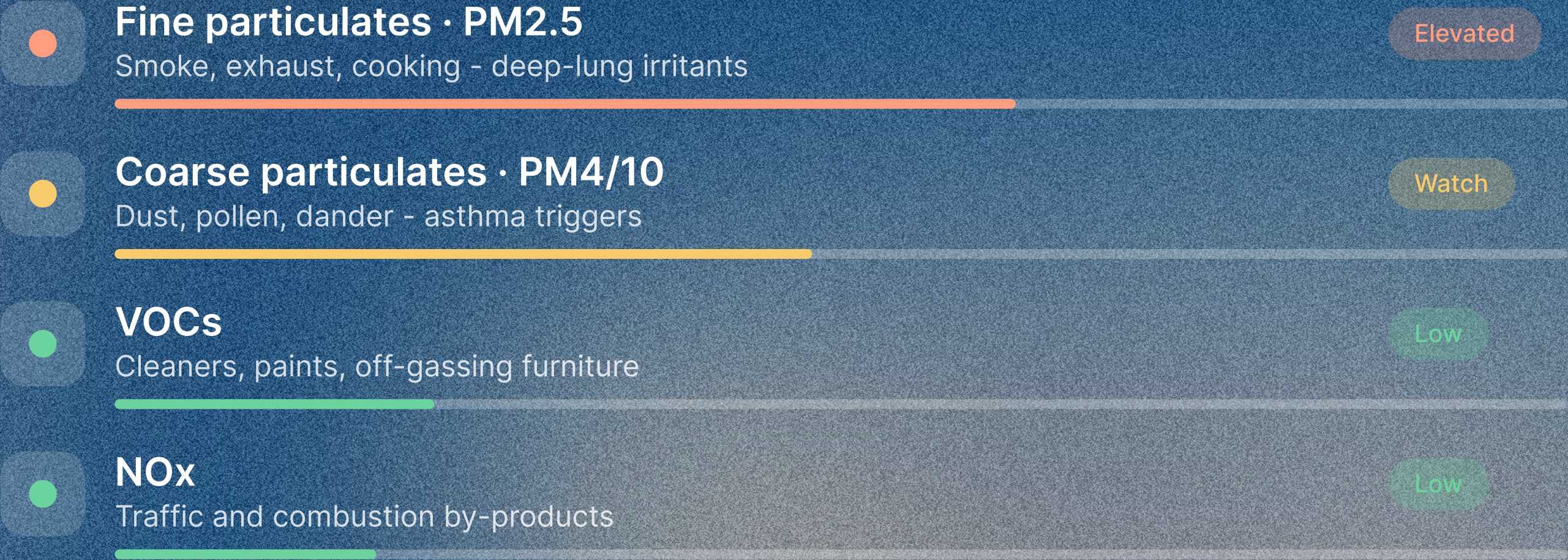
WHAT GOES INTO IT



Weighted for cognitive impact: CO₂ dominates, particulates and thermal comfort modulate. Personalizes over time as your history builds.

For the air your lungs care about.

Included with every Perch



Built for people with asthma, allergies and respiratory sensitivities - awareness and everyday decisions. Perch is not a medical device and does not diagnose or treat any condition.

Find the trigger. Stop guessing.

100M+ Americans live with conditions their environment can set off. They keep diaries, cut foods, guess. The one variable nobody measures is the environment they actually lived in - Perch measures it continuously.

Log in one tap



Migraine. Flare. Wheeze. A symptom log designed for bad days: one tap, timestamped, done. The Perch bird checks in - you don't fill out forms.

Cross-reference automatically

Every symptom is matched against your continuous exposure record - CO₂, PM, VOC, NO_x, temperature, humidity and barometric pressure - across home, car, office and outdoors.

Surface the pattern

After enough data: "Your migraines cluster on pressure-drop days with high VOC exposure." A personal trigger map you can act on - and bring to your doctor.

WHO IT'S FOR

- Allergies & asthma
- Migraine & cluster headache
- Eczema & rashes
- MS heat sensitivity
- Arthritis flares
- Patterns, not diagnoses

One device. Every signal.

Perch - \$179

The device

Built on the Sensirion SEN66

- ✓ CO₂, temperature and humidity - plus PM1 / PM2.5 / PM4 / PM10, VOC, NOx and barometric pressure
- ✓ Pocket clip-on · e-ink display · multi-day battery
- ✓ Carabiner loop; pairs & syncs over BLE

Alpha prototype

Working today

Built on the Sensirion SCD-41

- ✓ CO₂, temperature and humidity pairing & syncing with the app
- ✓ Proves the platform: BLE sync, scores and the app pipeline
- ✓ Production design swaps in the SEN66 - architecture designed



The beta device. Every part serviceable.

82 × 86 × 39 mm body · 2.13" e-paper display · Ø25 mm crown · RGB light pipe

Sensirion SEN66 (PM · CO₂ · VOC · NO_x · temp · RH), Bosch barometer and GNSS with flex antenna - the full exposure record in one unit

4400 mAh 1S2P pack - multi-day duty-cycled runtime

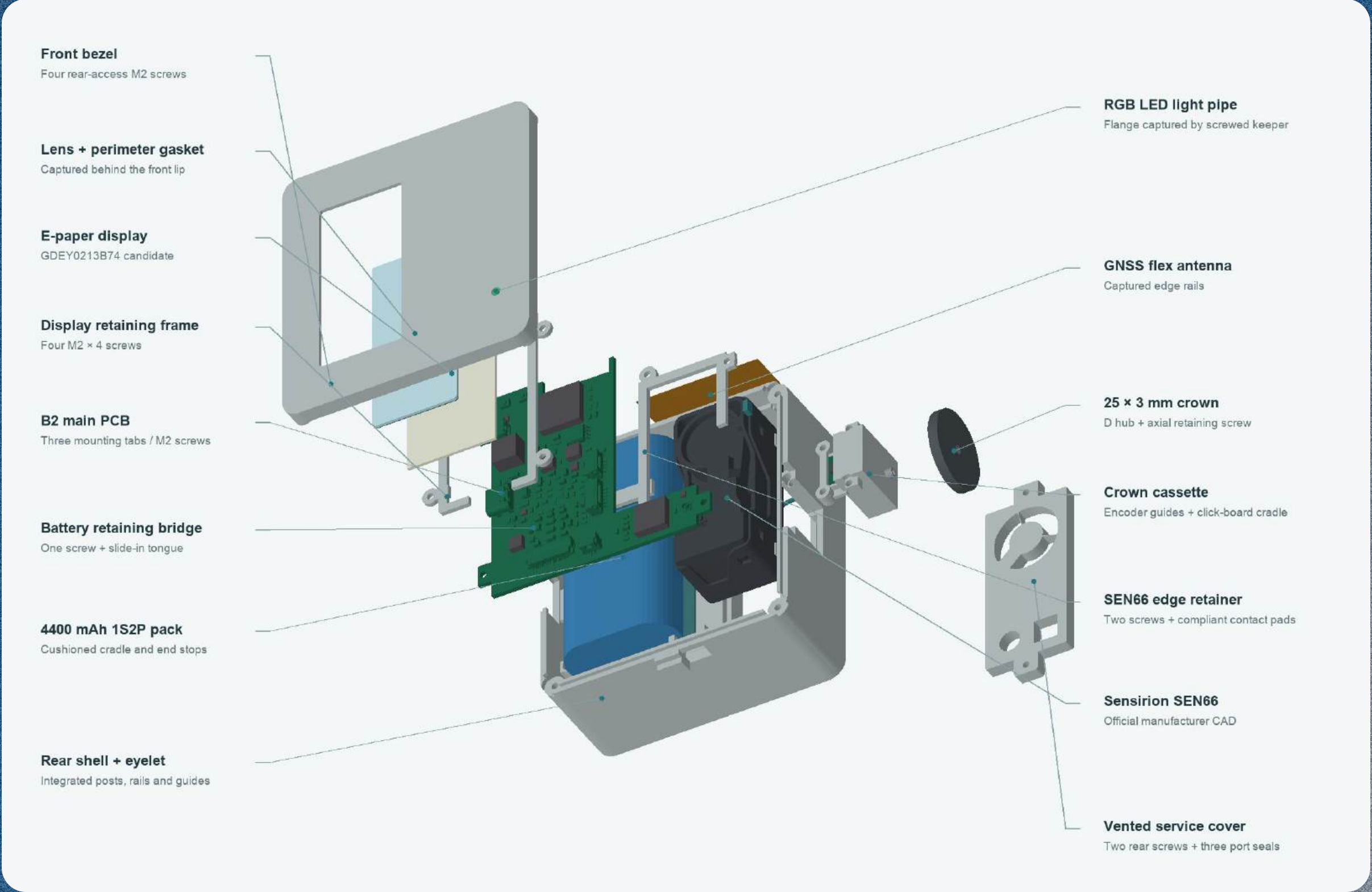
Screw-closure case with removable retainers: battery, sensor and PCB all replaceable. Beta units get repaired overnight, not returned.



Fit prototype: print coupons + one fit assembly next · purchased-part dimensions and crown mechanism still to validate · not yet a PCB release.

What's inside - and what it costs.

COMPONENT	PROTO	AT 1K
Sensirion SEN66 - PM, CO ₂ , VOC, NO _x , temp, RH	\$61	\$42
Bosch barometric sensor	\$2	\$1
GNSS module + flex antenna	\$8	\$4
ESP32-S3 module - BLE + Wi-Fi	\$10	\$2.50
2.13" e-paper display	\$12	\$5
Crown encoder + click board	\$4	\$2
4400 mAh 1S2P battery pack	\$12	\$5
Power management + USB-C	\$3	\$2
PCB / PCBA	\$15	\$9
Enclosure + hardware (printed → molded)	\$12	\$4
Assembly & test	-	\$10
TOTAL	~\$139	~\$87



Distributor pricing, 2026 (project cost model). Molded-enclosure figure excludes ~\$10K tooling. The SEN66 is ~half the BOM - and the reason one device carries every signal. At \$179 retail: ~50% gross margin at 1,000 units.

What's in your air – and what it does at high concentrations.

- **Carbon dioxide · CO₂**

Decision-making and complex thinking measurably slow; drowsiness at the levels cars (4,000+ ppm) and bedrooms (2,000+) routinely reach.

- **Fine particulates · PM_{2.5}**

Smoke, exhaust and cooking aerosols penetrate deep into the lungs – linked to reduced attention and long-term cardiovascular harm.

- **Coarse particulates · PM₁₀**

Dust, pollen and dander irritate airways – the classic asthma and allergy triggers.

- **Volatile organic compounds**

Paints, cleaners and off-gassing furniture cause headaches, fatigue and irritation in sealed rooms.

- **Nitrogen oxides · NO_x**

Traffic and gas-stove combustion inflame airways – hitting sensitive lungs hardest.

What else is around you - and what it does to how you feel.

- **Barometric pressure · hPa**

Pressure drops and swings are among the most-cited migraine and headache triggers - roughly half of sufferers report weather sensitivity (Mukamal et al., 2009).

- **Sound pollution · dB**

Chronic noise above ~55 dB disturbs sleep and raises stress and cardiovascular risk - WHO ranks traffic noise among the top environmental health burdens in Europe.

- **Temperature · °C**

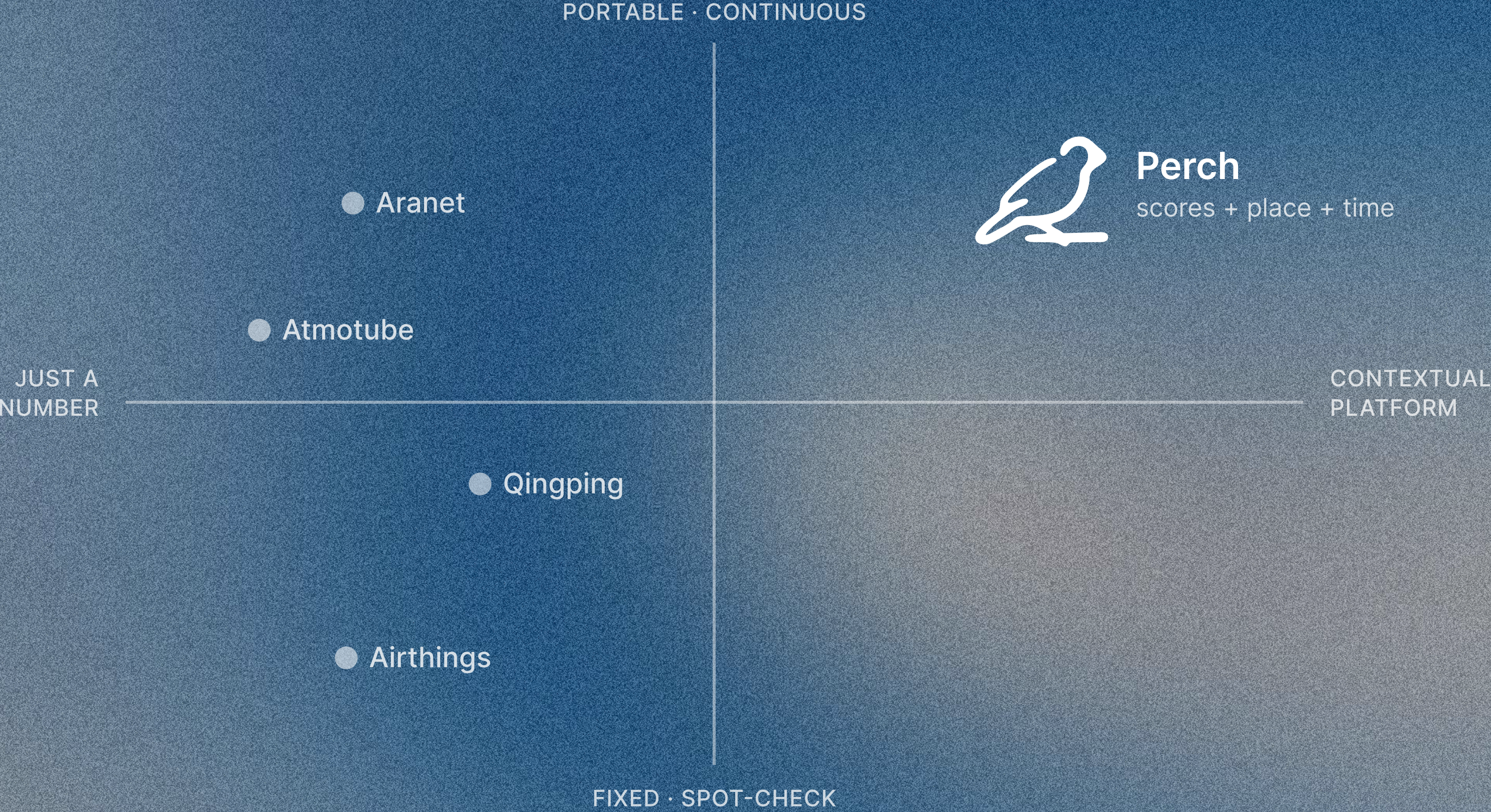
Heat measurably degrades focus and fragments sleep - bedrooms sleep best near 18 °C - and small rises temporarily worsen MS symptoms (Uhthoff's phenomenon).

- **Humidity · %RH**

Below ~40% RH airways and skin dry out - eczema and irritation flare; above ~60% dust mites and mould thrive - the classic allergy triggers.

Together with air quality, these complete the exposure record Perch cross-references against your symptoms and scores.

Everyone sells the number.
We own the meaning.



The sensor is a commodity. The score and data layer is the moat.



Contextual

Air mapped to exact place and moment - not a number floating in space.



Longitudinal

Trends over weeks and months; every reading deepens the personal baseline.



Research-based scores

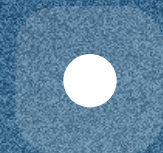
CognAir, SleepAir, FitAir, HealthAir - proprietary, patent-pending, and they personalize with use.



THE OURA PARALLEL

Oura didn't invent the heart-rate sensor - it won on the daily score and habit loop. Perch runs that exact playbook for air, with a device whose full sensor suite feeds the scores rich signal from day one.

The exposure map nobody else has.



What we collect

Longitudinal CO₂, PM, VOC, NO_x, temperature, humidity and pressure readings - indoors and outdoors, tied to place, time and symptoms.



Privacy first

Anonymized and aggregated; members own their data and opt in to sharing. Never sold at the individual level.



Who buys

HVAC & smart-home OEMs, insurers & real estate, and research & public health - where symptom-tagged exposure data is the rarest layer.



THE BREEZOMETER PARALLEL

Google paid \$200M+ for BreezoMeter - outdoor air modeled from public stations. Perch measures real exposure wherever members go: indoors, where 90% of life happens, and outdoors, street by street. Nobody owns that map.

The Perch Map API.

Live exposure maps

Block-by-block, hour-by-hour - indoors and out.

Slice any signal

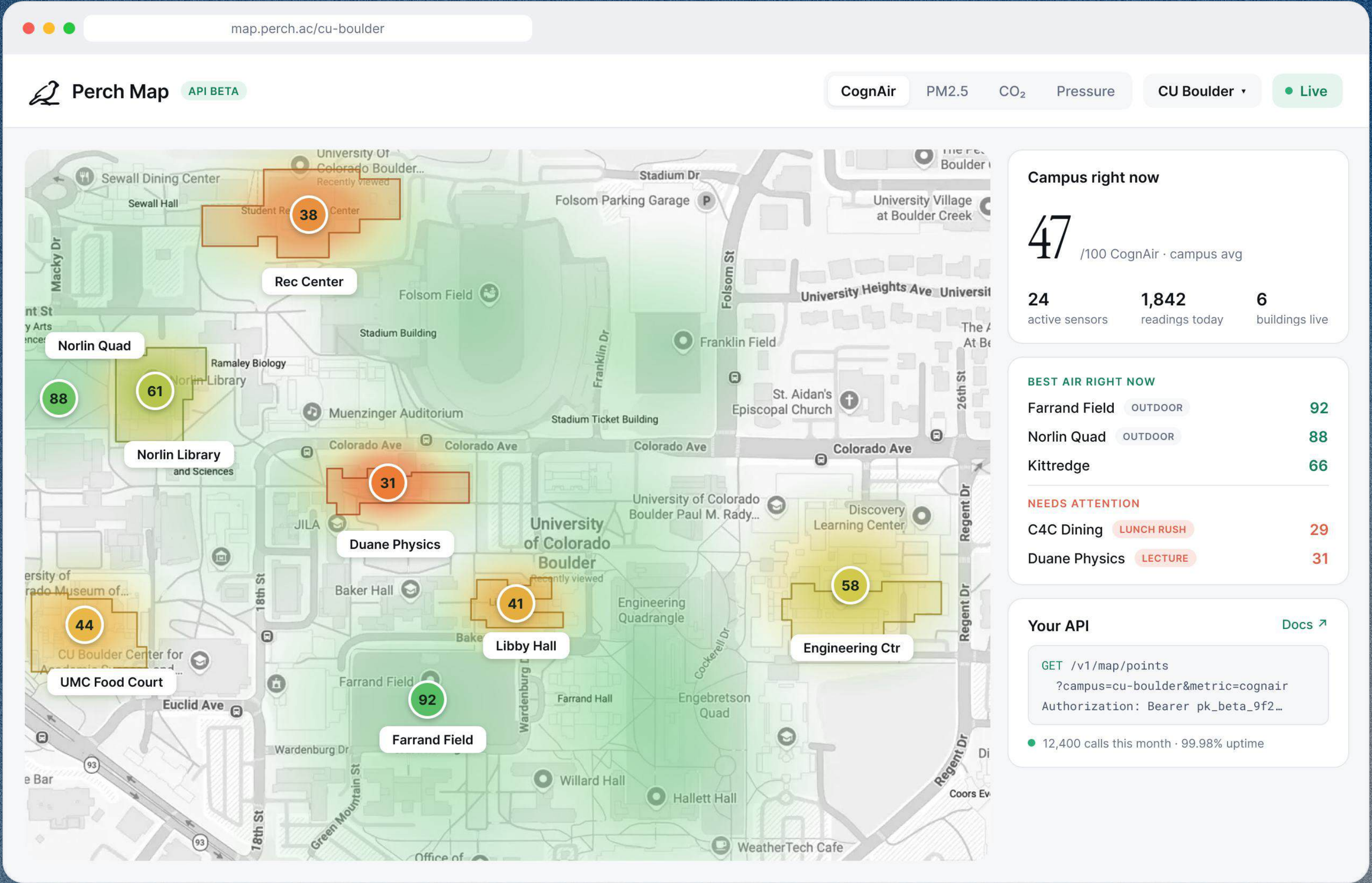
Filter by pollutant, pressure, time window and place type.

Privacy-first by design

Aggregated and anonymized - never individual-level.

Dashboard + REST API

For campuses, HVAC & smart-home OEMs, insurers and researchers.



Built to be checked daily - not drawered.

Air monitors have a graveyard problem: bought, glanced at for three weeks, forgotten. We designed Perch against that failure mode from day one.

An AI companion, not a dashboard



The Perch bird is interactive - it notices your air, talks to you in plain language, and hands you one tip at a time. A character you check in on (the Duolingo effect), not a chart you abandon.

A daily score ritual

CognAir in the morning, SleepAir when you wake, HealthAir when it matters. Scores change every day because your air does - the same habit loop that keeps Oura and Whoop opened daily.

A device worth glancing at

Perch's large display makes the device itself a touchpoint - live score, bird and tips at a glance. Screenless dongles like Atmotube disappear into pockets; Perch stays part of your day.

THE LOOP



We stay focused: sensors and the platform.

- **Months 0–3** Perch alpha polish · SEN66 production prototype built · proprietary score data from real bedrooms and commutes
- **Months 3–9** Campus betas - 100+ users across CU Boulder, UCSD and more; ~25 per campus is enough to map it · retention and score validation · first campus-wide exposure maps
- **Months 9–12** Host institutions convert to our first paying data customers - hardware validated and revenue proven in one motion; seed closes on that proof
- **Months 12–18** Paid launch: Perch \$179 · lighthouse fleet and workplace pilots · data licensing expands beyond campuses

THE PLATFORM EXTENDS - WITHOUT BUILDING APPLIANCES

Matter & HomeKit

Perch scores become triggers in the smart-home standard - your purifier, ERV or thermostat reacts automatically.

IFTTT & webhooks

Anyone can wire Perch to what they already own: fans, windows, alerts, dashboards - no new hardware from us.

Open API & partners

Air-quality data as a service for HVAC, fleet-safety and wellness platforms - distribution without manufacturing.

The evidence behind the claims.

COGNITION & PERFORMANCE (PROBLEM, COGNAIR, SCIENCE)

Du, B., Tandoc, M. C., Mack, M. L., & Siegel, J. A. (2020). Indoor CO₂ concentrations and cognitive function: A critical review. *Indoor Air*, 30(6), 1067–1082. doi:10.1111/ina.12706

Thompson, R., et al. (2023). Air pollution and human cognition: A systematic review and meta-analysis. *Science of The Total Environment*, 859, 160234. doi:10.1016/j.scitotenv.2022.160234

Clifford, A., Lang, L., Chen, R., Anstey, K. J., & Seaton, A. (2016). Exposure to air pollution and cognitive functioning across the life course – A systematic literature review. *Environmental Research*, 147, 383–398. doi:10.1016/j.envres.2016.01.018

Chandra, M., et al. (2022). Air pollution and cognitive impairment across the life course in humans: A systematic review. *Int. J. Environmental Research and Public Health*, 19(3), 1405. doi:10.3390/ijerph19031405

de Prado Bert, P., et al. (2018). The effects of air pollution on the brain: A review of studies interfacing environmental epidemiology and neuroimaging. *Current Environmental Health Reports*, 5(3), 351–364. doi:10.1007/s40572-018-0209-9

Peters, R., et al. (2019). Air pollution and dementia: A systematic review. *Journal of Alzheimer's Disease*, 70(s1), S145–S163. doi:10.3233/JAD-180631

Sunyer, J., et al. (2017). Traffic-related air pollution and attention in primary school children: Short-term association. *Epidemiology*, 28(2), 181–189. doi:10.1097/EDE.0000000000000603

Ebenstein, A., Lavy, V., & Roth, S. (2016). The long-run economic consequences of high-stakes examinations: Evidence from transitory variation in pollution. *American Economic Journal: Applied Economics*, 8(4), 36–65. doi:10.1257/app.20150213

SLEEP & FITNESS (SLEEPAIR, FITAIR)

Wang, M., Cho, Y., & Li, J. (2025). Air pollution and sleep health in middle-aged and older adults: A systematic review and meta-analysis. *Sleep Medicine Reviews*, 83, 102136. doi:10.1016/j.smr.2025.102136

Hung, A., Nelson, H., & Koehle, M. S. (2022). The acute effects of exercising in air pollution: A systematic review of randomized controlled trials. *Sports Medicine*, 52(1), 139–164. doi:10.1007/s40279-021-01544-4

Rundell, K. W. (2012). Effect of air pollution on athlete health and performance. *British Journal of Sports Medicine*. doi:10.1136/bjsports-2011-090823

The evidence behind the claims.

ENVIRONMENTAL TRIGGERS OF CHRONIC CONDITIONS (PROBLEM, HEALTHAIR, SYMPTOM TRACKING)

Mukamal, K. J., Wellenius, G. A., Suh, H. H., & Mittleman, M. A. (2009). Weather and air pollution as triggers of severe headaches. *Neurology*, 72(10), 922–927. doi:10.1212/01.wnl.0000344152.56020.94

Teekaput, C., et al. (2025). Impact of air pollution on the clinical exacerbation of central demyelinating disease: 10-year data from the Northern Thailand MS and NMOSD registry. *Multiple Sclerosis and Related Disorders*, 94, 106266. doi:10.1016/j.msard.2025.106266

Adami, G., et al. (2021). Association between environmental air pollution and rheumatoid arthritis flares. *Rheumatology*. doi:10.1093/rheumatology/keab049

Chang, K.-H., et al. (2016). Air pollution exposure increases the risk of rheumatoid arthritis: A longitudinal and nationwide study. *Environment International*, 94, 495–499. doi:10.1016/j.envint.2016.06.008

Zhang, J., et al. (2023). Association of combined exposure to ambient air pollutants, genetic risk, and incident rheumatoid arthritis: A prospective cohort study in the UK Biobank. *Environmental Health Perspectives*, 131. doi:10.1289/ehp10710

Liu, J.-F., et al. (2020). PM2.5 facilitates IL-6 production in human osteoarthritis synovial fibroblasts via ASK1 activation. *Journal of Cellular Physiology*, 236, 2205–2213. doi:10.1002/jcp.30009

INDOOR POLLUTANTS (SCIENCE, HEALTHAIR)

Liu, N., et al. (2022). Health effects of exposure to indoor volatile organic compounds from 1980 to 2017: A systematic review and meta-analysis. *Indoor Air*, 32(5), e13038. doi:10.1111/ina.13038